HybridKernel: A Profiler-Driven Boundary-Fusion Gate for Hybrid Attention/SSM Serving

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Abstract

Hybrid attention/state-space language models invite a systems question: do attention-to-SSM layer boundaries create avoidable conversion, materialization, launch, or locality overhead that a fused boundary operator could remove? We report a Mac-feasible pre-GPU gate for this question. Architecture maps show large boundary-crossing activation streams, but a runtime audit weakens the broad novelty claim because current vLLM hybrid SSM support already handles important state-layout and transfer paths. A threshold model says Granite 4.0 H needs roughly 25% genuinely avoidable boundary traffic at 60% recovery to clear a 3% proxy gain; Qwen3-Next needs 10.4% but is less directly matched to the Granite Mamba2 hypothesis. We therefore do not claim a kernel speedup. The contribution is a reproducible profiler gate, a fixed-request vLLM driver, and a pre-registered profiler-analysis parser with a promote/kill criterion for future native NVIDIA measurements.

1 Hypothesis

HybridKernel tests whether adjacent attention and SSM layers expose boundary-local overhead not already removed by serving systems. Candidate costs include layout conversion, residual-stream materialization, host launch gaps, cache-locality loss, or redundant memory traffic. The hypothesis is deliberately narrow: a useful kernel must compose with existing vLLM hybrid mechanisms rather than replace them.

2 Current Evidence

Artifact	Result	Decision
Architecture map	Granite has 8 boundaries and 20.0% boundary stream fraction; Qwen3-Next has 23 inferred boundaries and 47.9%.	Inspectable, not speed evidence.
Runtime audit	vLLM already handles hybrid state layout and transfer paths.	Broad novelty weakened.
Threshold model	Granite needs about 25.0% avoidable boundary traffic at 60% recovery for a 3% proxy gain.	Mac kernel work not justified.
Profiler driver	Fixed-request OpenAI-compatible driver dry-runs locally.	Native gate is runnable.
Analysis parser	JSON profiler summaries are reduced into avoidable-boundary share and recoverable-gain upper bound.	Decision rule is pre-registered.

Table 1: HybridKernel evidence before native GPU profiling.

3 Profiler Gate

The next experiment is a native NVIDIA/vLLM run with Nsight Systems and Nsight Compute. It must show a distinct boundary-local overhead attributable to attention/SSM transitions, not warmup, graph capture, batching, or unrelated kernels. Promotion requires at least three repeated runs whose recoverable-gain upper bound clears 3%. The analysis parser computes

$$\text{gain}_{ub} = \frac{\max(\text{boundary ms} - \text{matched control ms}, 0)}{\text{total step ms}} \times \text{recoverable fraction}.$$

If repeated native summaries show less than 1% recoverable gain, the branch should be killed or shelved.

4 Claim Boundary

Allowed now: HybridKernel is a weakly alive profiler-driven systems branch with a runnable native measurement plan. Not allowed: throughput, latency, HBM, energy, occupancy, or vLLM speedup claims. If native profiling does not reveal separable boundary overhead, the branch should be killed.

5 Artifacts

- `experimental/hybridkernel/phase2/nvidia_vllm_profiler_runbook.md`
- `experimental/hybridkernel/phase2/profiler_driver.py`
- `experimental/hybridkernel/phase2/analyze_profiler_metrics.py`
- `experimental/hybridkernel/phase2/profiler_analysis_gate.md`
- `experimental/hybridkernel/phase2/pre_gpu_threshold_model.md`
- `experimental/hybridkernel/paper/colm_workshop_scaffold.md`